

Image-Only Evaluation Metrics

V3 Multimodal Fake-News Detector - aux head (binary task)

Table 1. Test configuration.

Field	Value
Evaluation date	2026-05-21
Model	V3 fusion module (Qwen2-7B variant)
Image encoder	UnivFD (ResNet-50, 768-d)
Preprocessing	Patch-DCT 8x8, YCbCr Y-channel, log + min-max
Classifier head	V3 aux head, Linear(768, 2), BCE-trained
Hardware	NVIDIA GPU, CUDA
Test split	2475 samples (495 per original class)
Task	Binary - image is AI/tampered
Positive label	original class in {Manipulated, Fully-Fabricated}
Decision threshold	0.50

Table 2. Binary classification metrics on the test split.

Metric	Value
Accuracy	0.6562
F1 (fake-image class)	0.2581
F1-macro	0.5171
Precision (fake)	0.9427
Recall (fake)	0.1495
AUC-ROC	0.7576
Samples evaluated	2475
Samples failed	0

Table 3. Confusion matrix (rows = ground truth, columns = prediction).

	Predicted: Real	Predicted: Fake
Actual: Real (n = 1485)	1476	9
Actual: Fake (n = 990)	842	148

Table 4. Per-original-class breakdown.

#	Class	n	Image truly fake	Mean P(fake)	% predicted fake	Slice accuracy
0	Real	495	no	0.281	0.4%	99.6%
1	Out-of-Context	495	no	0.290	0.8%	99.2%

2	Manipulated	495	yes	0.300	0.8%	0.8%
3	AI-Text	495	no	0.277	0.6%	99.4%
4	Fully Fabricated	495	yes	0.475	29.1%	29.1%

Table 5. Reference - full multimodal V3 on the same test split.

Model variant	Classes	F1-macro	AUC-ROC
V3 RoBERTa rev 12 (canonical)	5	0.5391	approx 0.85
V3 Qwen2-7B rev 11 (deployed)	5	0.5377	approx 0.84
Image-only aux head (this report)	2	0.5171	0.7576
Phase-2 baseline (3-class)	3	0.4912	approx 0.77

Table 6. Reference - zero-shot transfer to MMFakeBench.

Model variant	F1-macro	AUC-macro
V3 Qwen2-7B rev 9	0.3832	0.8168
V3 RoBERTa rev 12	0.3832	0.8168
Phase-2 baseline (3-class)	0.3083	0.7406